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Display with Collimating Optics Including an Annular Mirror

Abstract

An electronic device may include a display module that generates light and an optical system that redirects the light towards an eye box. The system may include an input coupler on a waveguide and collimating optics that collimate the light between the light exiting a display module and the light reaching the input coupler. The collimating optics may include a first mirror with a convex surface on an opposing side of the waveguide as the display module. The collimating optics may include a second mirror with an annular footprint on the same side of the waveguide as the display module. The second mirror may have a concave surface and may laterally surround the display module. The input coupler may have an annular footprint that laterally surrounds the first mirror.

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Background/Summary

[0001] This application claims the benefit of U.S. provisional patent application No. 63/558,522 filed Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] This relates generally to optical systems and, more particularly, to optical systems for displays.

[0003] Electronic devices may include displays that present images to a user's eyes. For example, devices such as virtual reality and augmented reality headsets may include displays with optical elements that allow users to view the displays.

[0004] It can be challenging to design devices such as these. If care is not taken, the components used in displaying content may be unsightly and bulky and may not exhibit desired levels of optical performance.

SUMMARY

[0005] A display system may include a waveguide having first and second opposing sides, a first mirror on the first side of the waveguide, a display module on the second side of the waveguide that produces image light, and a second mirror on the second side of the waveguide. The second mirror may have an annular footprint with a central opening and the display module and the first mirror may overlap the central opening.

[0006] A display system may include a waveguide having first and second opposing sides, a first mirror on the first side of the waveguide, a surface relief grating on the first side of the waveguide that has a first annular footprint with a first central opening that overlaps the first mirror, a display module on the second side of the waveguide that produces image light that passes through the first central opening to reach the first mirror, and a second mirror on the second side of the waveguide that receives the image light from the first mirror.

[0007] A display system may include a waveguide having first and second opposing sides, a display module that produces image light, an input coupler on the waveguide that couples the image light into the waveguide, and collimating optics that collimate the image light from the display module for the input coupler. The collimating optics may include a first mirror on the first side of the waveguide and a second mirror on the second side of the waveguide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram of an illustrative system having a display in accordance with some embodiments.

[0009] FIG. 2 is a top view of an illustrative optical system for a display having a waveguide with an input coupler in accordance with some embodiments.

[0010] FIG. 3 is a top view of an illustrative optical system for a display having collimating optics that include an annular mirror in accordance with some embodiments.

[0011] FIGS. 4A and 4B are side views of the display of FIG. 3 in accordance with some embodiments.

[0012] FIG. 5 is a top view of an illustrative optical system for a display having collimating optics that include concentric annular mirrors in accordance with some embodiments.

[0013] FIG. 6 is a side view of the concentric annular mirrors of FIG. 5 in accordance with some embodiments.

DETAILED DESCRIPTION

[0014] An illustrative system having a device with one or more near-eye display systems is shown

in FIG. 1. System **10** may be a head-mounted device having one or more displays such as near-eye displays **14** mounted within support structure (housing) **20**. Support structure **20** may have the shape of a pair of eyeglasses (e.g., supporting frames), may form a housing having a helmet shape, or may have other configurations to help in mounting and securing the components of near-eye displays **14** on the head or near the eye of a user. Near-eye displays **14** may include one or more display modules such as display modules **14A** and one or more optical systems such as optical systems **14B**. Display modules **14A** may be mounted in a support structure such as support structure **20**. Each display module **14A** may emit light **22** (image light) that is redirected towards a user's eyes at eye box **24** using an associated one of optical systems **14B**.

[0015] The operation of system **10** may be controlled using control circuitry **16**. Control circuitry **16** may include storage and processing circuitry for controlling the operation of system **10**. Circuitry **16** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **16** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code (instructions) may be stored on storage in circuitry **16** and run on processing circuitry in circuitry **16** to implement operations for system **10** (e.g., data gathering operations, operations involving the adjustment of components using control signals, image rendering operations to produce image content to be displayed for a user, etc.).

[0016] System **10** may include input-output circuitry such as input-output devices **12**. Input-output devices **12** may be used to allow data to be received by system **10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, or other electrical equipment) and to allow a user to provide head-mounted device **10** with user input. Input-output devices **12** may also be used to gather information on the environment in which system **10** (e.g., head-mounted device **10**) is operating. Output components in devices **12** may allow system **10** to provide a user with output and may be used to communicate with external electrical equipment. Input-output devices **12** may include sensors and other components **18** (e.g., image sensors for gathering images of real-world object that are digitally merged with virtual objects on a display in system **10**, accelerometers, depth sensors, light sensors, haptic output devices, speakers, batteries, wireless communications circuits for communicating between system **10** and external electronic equipment, etc.).

[0017] Display modules **14A** may include reflective displays (e.g., liquid crystal on silicon (LCOS) displays, digital-micromirror device (DMD) displays, or other spatial light modulators), emissive displays (e.g., micro-light-emitting diode (uLED) displays, organic light-emitting diode (OLED) displays, laser-based displays, etc.), or displays of other types. Light sources in display modules **14A** may include uLEDs, OLEDs, LEDs, lasers, combinations of these, or any other desired light-emitting components.

[0018] Optical systems **14B** may form lenses that allow a viewer (see, e.g., a viewer's eyes at eye box **24**) to view images on display(s) **14**. There may be two optical systems **14B** (e.g., for forming left and right lenses) associated with respective left and right eyes of the user. A single display **14** may produce images for both eyes or a pair of displays **14** may be used to display images. In configurations with multiple displays (e.g., left and right eye displays), the focal length and positions of the lenses formed by components in optical system **14B** may be selected so that any gap present between the displays will not be visible to a user (e.g., so that the images of the left and right displays overlap or merge seamlessly).

[0019] If desired, optical system **14B** may contain components (e.g., an optical combiner, etc.) to allow real-world image light from real-world images or objects **25** to be combined optically with virtual (computer-generated) images such as virtual images in image light **22**. In this type of

system, which is sometimes referred to as an augmented reality system, a user of system **10** may view both real-world content and computer-generated content that is overlaid on top of the real-world content. Camera-based augmented reality systems may also be used in device **10** (e.g., in an arrangement which a camera captures real-world images of object **25** and this content is digitally merged with virtual content at optical system **14B**).

[0020] System **10** may, if desired, include wireless circuitry and/or other circuitry to support communications with a computer or other external equipment (e.g., a computer that supplies display **14** with image content). During operation, control circuitry **16** may supply image content to display **14**. The content may be remotely received (e.g., from a computer or other content source coupled to system **10**) and/or may be generated by control circuitry **16** (e.g., text, other computer-generated content, etc.). The content that is supplied to display **14** by control circuitry **16** may be viewed by a viewer at eye box **24**.

[0021] FIG. **2** is a top view of an illustrative display **14** that may be used in system **10** of FIG. **1**. As shown in FIG. **2**, near-eye display **14** may include one or more display modules such as display module **14A** and an optical system such as optical system **14B**. Optical system **14B** may include optical elements such as one or more waveguides **26**. Waveguide **26** may include one or more stacked substrates (e.g., stacked planar and/or curved layers sometimes referred to herein as waveguide substrates) of optically transparent material such as plastic, polymer, glass, etc.

[0022] If desired, waveguide **26** may also include one or more layers of holographic recording media (sometimes referred to herein as holographic media, grating media, or diffraction grating media) on which one or more diffractive gratings are recorded (e.g., holographic phase gratings, sometimes referred to herein as holograms). A holographic recording may be stored as an optical interference pattern (e.g., alternating regions of different indices of refraction) within a photosensitive optical material such as the holographic media. The optical interference pattern may create a holographic phase grating that, when illuminated with a given light source, diffracts light to create a three-dimensional reconstruction of the holographic recording. The holographic phase grating may be a non-switchable diffractive grating that is encoded with a permanent interference pattern or may be a switchable diffractive grating in which the diffracted light can be modulated by controlling an electric field applied to the holographic recording medium. Multiple holographic phase gratings (holograms) may be recorded within (e.g., superimposed within) the same volume of holographic medium if desired. The holographic phase gratings may be, for example, volume holograms or thin-film holograms in the grating medium. The grating media may include photopolymers, gelatin such as dichromated gelatin, silver halides, holographic polymer dispersed liquid crystal, or other suitable holographic media.

[0023] Diffractive gratings on waveguide **26** may include holographic phase gratings such as volume holograms or thin-film holograms, meta-gratings, or any other desired diffractive grating structures. The diffractive gratings on waveguide **26** may also include surface relief gratings formed on one or more surfaces of the substrates in waveguides **26**, gratings formed from patterns of metal structures, etc. The diffractive gratings may, for example, include multiple multiplexed gratings (e.g., holograms) that at least partially overlap within the same volume of grating medium (e.g., for diffracting different colors of light and/or light from a range of different input angles at one or more corresponding output angles).

[0024] Optical system **14B** may include collimating optics such as collimating optics **34**. Collimating lens **34** may include one or more lens elements and/or mirrors that help direct image light **22** towards waveguide **26**. If desired, display module **14A** may be mounted within support structure **20** of FIG. **1** while optical system **14B** may be mounted between portions of support structure **20** (e.g., to form a lens that aligns with eye box **24**). Other mounting arrangements may be used, if desired.

[0025] As shown in FIG. **2**, display module **14A** may generate light **22** associated with image content to be displayed to eye box **24**. Light **22** may be collimated using collimating optics **34**.

Optical system **14B** may be used to present light **22** output from display module **14A** to eye box **24**. [0026] Optical system **14B** may include one or more optical couplers such as input coupler **28**, cross-coupler **32**, and output coupler **30**. In the example of FIG. 2, input coupler **28**, cross-coupler **32**, and output coupler **30** are formed at or on waveguide **26**. Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be completely embedded within the substrate layers of waveguide **26**, may be partially embedded within the substrate layers of waveguide **26**, may be mounted to waveguide **26** (e.g., mounted to an exterior surface of waveguide **26**), etc.

[0027] The example of FIG. 2 is merely illustrative. One or more of these couplers (e.g., cross-coupler **32**) may be omitted. Optical system **14B** may include multiple waveguides that are laterally and/or vertically stacked with respect to each other. Each waveguide may include one, two, all, or none of couplers **28**, **32**, and **30**. Waveguide **26** may be at least partially curved or bent if desired.

[0028] Waveguide **26** may guide light **22** down its length via total internal reflection. Input coupler **28** may be configured to couple light **22** from display module **14A** (lens **34**) into waveguide **26**, whereas output coupler **30** may be configured to couple light **22** from within waveguide **26** to the exterior of waveguide **26** and towards eye box **24**. For example, display module **14A** may emit light **22** in direction +Y towards optical system **14B**. When light **22** strikes input coupler **28**, input coupler **28** may redirect light **22** so that the light propagates within waveguide **26** via total internal reflection towards output coupler **30** (e.g., in direction X). When light **22** strikes output coupler **30**, output coupler **30** may redirect light **22** out of waveguide **26** towards eye box **24** (e.g., back along the Y-axis). In scenarios where cross-coupler **32** is formed at waveguide **26**, cross-coupler **32** may redirect light **22** in one or more directions as it propagates down the length of waveguide **26**, for example.

[0029] Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be based on reflective and refractive optics or may be based on holographic (e.g., diffractive) optics. In arrangements where couplers **28**, **30**, and **32** are formed from reflective and refractive optics, couplers **28**, **30**, and **32** may include one or more reflectors (e.g., an array of micromirrors, partial mirrors, or other reflectors). In arrangements where couplers **28**, **30**, and **32** are based on holographic optics, couplers **28**, **30**, and **32** may include diffractive gratings (e.g., volume holograms, surface relief gratings, etc.).

[0030] FIG. 3 is a top view of an illustrative display **14** that includes collimating optics with reflective mirrors. As shown, display module **14A** is positioned on a first side of waveguide **26**. Display module **14A** emits light in the positive Y-direction through waveguide **26** to a mirror **48** that is positioned on a second, opposing side of waveguide **26**. Mirror **48** therefore overlaps display panel module **14A** in the Y-direction.

[0031] Mirror **48** has a surface with convex curvature (sometimes referred to as convex surface **48-C**) configured to reflect the light received from display module **14A** in the negative Y-direction. Surface **48-C** may be referred to as facing display module **14A**. After being reflected by convex surface **48-C**, the light may pass through the waveguide in the negative Y-direction towards mirror **44**. Mirror **44** laterally surrounds display module **14A** within the XZ-plane. In other words, mirror **44** has an annular footprint with a central opening and display module **14A** and mirror **48** overlap the central opening in the Y-direction. Display module **14A** may be referred to as being positioned within the central opening of mirror **44**. Mirror **44** may sometimes be referred to as a ring-shaped mirror or annular mirror.

[0032] Mirror **44** has a surface with concave curvature (sometimes referred to as concave surface **44-C**) configured to reflect the light received from mirror **48** in the positive Y-direction towards input coupler **28**. Surface **44-C** may be referred to as facing input coupler **28**. After being reflected by concave surface **44-C**, the light may pass through the waveguide in the positive Y-direction to input coupler **28**. In the example of FIG. 3, input coupler **28** is formed from a surface relief grating. Input coupler **28** laterally surrounds mirror **48** within the XZ-plane. In other words, input coupler has an annular footprint with a central opening and mirror **48** and display module **14A** overlap the

central opening in the Y-direction. Mirror **48** may be referred to as being positioned within the central opening of input coupler **28**. Input coupler **28** may sometimes be referred to as a ring-shaped input coupler or annular input coupler.

[0033] The input coupler **28** may receive light from mirror **44** and redirect the light so that the light propagates within waveguide **26** via total internal reflection (e.g., towards a cross coupler and/or output coupler **30** in the positive X-direction). If light is directed from input coupler **28** toward mirror **44**, light in total internal reflection will not re-interact with mirror **44** and will continue down waveguide **26** toward output coupler **30**.

[0034] Consider light rays **50-1** and **50-2** emitted by display module **14A**. Ray **50-1** is emitted by display module **14A** in the positive Y-direction. Ray **50-1** passes through the thickness of waveguide **26** a first time and is incident upon convex surface **48-C** of mirror **48**. Ray **50-1** subsequently passes through the thickness of waveguide **26** a second time and is incident upon concave surface **44-C** of mirror **44**. Ray **50-1** subsequently passes through the thickness of waveguide **26** a third time and is incident upon input coupler **28**. Ray **50-1** is subsequently redirected by input coupler **28** in the positive X-direction to propagate within waveguide **26** via total internal reflection.

[0035] Ray **50-2** is emitted by display module **14A** in the positive Y-direction. Ray **50-2** passes through the thickness of waveguide **26** a first time and is incident upon convex surface **48-C** of mirror **48**. Ray **50-2** subsequently passes through the thickness of waveguide **26** a second time and is incident upon concave surface **44-C** of mirror **44**. Rays **50-1** and **50-2** are incident upon concave surface **44-C** on first and second opposing sides of display module **14A**. Ray **50-2** subsequently passes through the thickness of waveguide **26** a third time and is incident upon input coupler **28**. Rays **50-1** and **50-2** are incident upon input coupler **28** on first and second opposing sides of mirror **48**. Ray **50-2** is subsequently redirected by input coupler **28** in the positive X-direction to propagate within waveguide **26** via total internal reflection.

[0036] Mirrors **44** and **48** may have a reflectance that is greater than 70%, greater than 80%, greater than 90%, greater than 95%, etc. The mirrors may be formed as coatings, films, or solid pieces. The mirrors may be attached or coated to one or more structures such as transparent structures **42** and **46** in FIG. 3. FIG. 3 shows an example where mirror **44** is coated on transparent structure **42**. Transparent structure **42** has the same footprint as mirror **44**. The transparent structure may be formed from transparent glass, plastic, etc. Transparent structure **42** may be attached to waveguide **26** (e.g., using optically clear adhesive). This example is merely illustrative and transparent structure may optionally be omitted if desired. When the transparent structure is omitted, there may be an air gap between waveguide **26** and mirror **44**. In FIG. 3, transparent structure **42** is interposed between mirror **44** and waveguide **26**. When the transparent structure is omitted, mirror **44** may be attached to an additional (transparent or opaque) structure that supports mirror **44** in a desired position adjacent to waveguide **26** (e.g., in the position shown in FIG. 3).

[0037] FIG. 3 shows an example where mirror **48** is coated on transparent structure **46**. Transparent structure **46** has the same footprint as mirror **48**. The transparent structure may be formed from transparent glass, plastic, etc. Transparent structure **46** may be attached to waveguide **26** (e.g., using optically clear adhesive). This example is merely illustrative and transparent structure may optionally be omitted if desired. When the transparent structure is omitted, there may be an air gap between waveguide **26** and mirror **48**. In FIG. 3, transparent structure **46** is interposed between mirror **48** and waveguide **26**. When the transparent structure is omitted, mirror **48** may be attached to an additional (transparent or opaque) structure that supports mirror **48** in a desired position adjacent to waveguide **26** (e.g., in the position shown in FIG. 3).

[0038] In the example of FIG. 3, transparent structure **42** has a convex surface that conforms to concave surface **44-C** and a planar surface that is parallel to waveguide **26**. This example is merely illustrative. Transparent structure **42** may, in general, have surfaces with any desired curvature. As one specific example, transparent structure **42** may have a convex surface that conforms to concave

surface **44-C** and an opposing concave surface. Transparent structure **42** may refract light from display module **14A** to help collimate the light for input coupler **28**.

[0039] Similarly, in the example of FIG. **3**, transparent structure **46** has a concave surface that conforms to convex surface **48-C** and a planar surface that is parallel to waveguide **26**. This example is merely illustrative. Transparent structure **46** may, in general, have surfaces with any desired curvature. As one specific example, transparent structure **46** may have a concave surface that conforms to convex surface **48-C** and an opposing convex surface. Transparent structure **46** may refract light from display module **14A** to help collimate the light for input coupler **28**. Either mirror **44** or mirror **48** may optionally be a Mangin mirror.

[0040] Waveguide **26** may optionally be encased in low-index cladding layers **52**. The low-index cladding layers may be formed from a material having an index of refraction that is less than 1.3, less than 1.2, less than 1.1, etc. In contrast, the index of refraction of waveguide **26** may be greater than 1.8, greater than 2.0, etc. The difference in index of refraction between waveguide **26** and low-index layers **52** may therefore be greater than 0.5, greater than 0.7, greater than 0.9, etc. The low-index cladding layers may optionally overlap and conform to a surface relief grating that defines input coupler **28**. The low-index cladding layers may be interposed between waveguide **26** and mirror **48**, may be interposed between waveguide **26** and transparent structure **46**, may be interposed between waveguide **26** and mirror **44**, and/or may be interposed between waveguide **26** and transparent structure **42**.

[0041] FIG. **4A** is a side view of display **14** from a perspective looking in the positive Y-direction. FIG. **4A** shows how display module **14A** overlaps waveguide **26** in the Y-direction. Mirror **44** also overlaps waveguide **26** in the Y-direction. As shown in FIG. **4A**, mirror **44** has an annular footprint with a central opening that overlaps display module **14A**. Said another way, mirror **44** completely laterally surrounds display module **14A**.

[0042] FIG. **4B** is a side view of display **14** from a perspective looking in the negative Y-direction. FIG. **4B** shows how mirror **48** overlaps waveguide **26** in the Y-direction. Input coupler **28** also overlaps waveguide **26** in the Y-direction. As shown in FIG. **4B**, input coupler **28** has an annular footprint with a central opening that overlaps mirror **48**. Said another way, input coupler **28** completely laterally surrounds mirror **48**. The footprints of display module **14A** and mirror **48** may overlap in the Y-direction. The footprints of input coupler **28** and mirror **44** may overlap in the Y-direction.

[0043] In the example of FIGS. **3**, **4A**, and **4B**, display module **14A** and annular mirror **44** are formed on the eye-box-side of waveguide **26**. In other words, light is ultimately output from waveguide **26** towards eye box **24** in the negative Y-direction (as shown in FIG. **2**). In FIG. **3**, display panel **14A** and annular mirror **44** are on the same side of waveguide **26** as eye box **24**. This example is merely illustrative. If desired, the arrangement may be flipped with display panel **14A** and annular mirror **44** instead being formed on the opposite side of waveguide **26** as eye box **24**.

[0044] As shown in FIGS. **4A** and **4B**, waveguide **26** may have a central region **62**, a first peripheral region **64**, and a second peripheral region **66**. Peripheral region **64** may sometimes be referred to herein as temple region **64** (e.g., because region **64** sits at or adjacent to the user's temple when the user is viewing the display). Peripheral region **66** may sometimes be referred to herein as nasal region **66** (e.g., because region **66** sits at or adjacent to the user's nose when the user is viewing the display).

[0045] In the example of FIGS. **3**, **4A**, and **4B**, display module **14A**, mirror **44**, mirror **48**, and input coupler **28** overlap the temple region of waveguide **26**. This example is merely illustrative. Display module **14A**, mirror **44**, mirror **48**, and input coupler **28** may instead overlap the nasal region of waveguide **26** if desired.

[0046] In the example of FIG. **3**, one annular mirror **44** surrounds display module **14A**. This example is merely illustrative. In another possible arrangement, shown in FIG. **5**, multiple annular mirrors may surround display module **14A**. As shown in FIG. **5**, a first annular mirror **44-1** has a

central opening that overlaps display module **14A**, a second annular mirror **44-2** has a central opening that overlaps display module **14A** and annular mirror **44-1**, a third annular mirror **44-3** has a central opening that overlaps display module **14A** and annular mirrors **44-1** and **44-2**, and a fourth annular mirror **44-4** has a central opening that overlaps display module **14A** and annular mirrors **44-1**, **44-2**, and **44-3**.

[0047] Each one of annular mirrors **44-1**, **44-2**, **44-3**, and **44-4** may have any of the same properties as described in connection with mirror **44** in FIG. 3. Each annular mirror may have a respective curvature and/or angle relative to the XZ-plane. Including multiple mirrors with different curvature and/or angles relative to the XZ-plane may provide additional degrees of freedom with which to control the collimation of light provided by display module **14A** to input coupler **28**.

[0048] In FIG. 5, a planar mirror **48** is positioned on waveguide **26** opposite annular mirrors **44**. Input coupler **28** laterally surrounds planar mirror **48**. FIG. 5 shows rays **50-1** and **50-2** being emitted by display module **14A** and reflecting off mirrors **44-1**, **44-2**, **44-3**, and **44-4** in that order. Planar mirror **48** reflects light multiple times as the light passes from display module **14A** to input coupler **28**. The light is ultimately received and redirected by input coupler **28** (as shown in connection with FIG. 3). The function of input coupler **28** is the same in FIG. 5 as in FIG. 3 and will not be discussed further in connection with FIG. 5.

[0049] The example of mirror **48** being planar is merely illustrative. If desired, mirror **48** may optionally have annular regions with curvature. Each annular region of mirror **48** may be configured to direct incident light to a respective one of mirrors **44**.

[0050] In FIG. 5, annular mirrors **44-1**, **44-2**, **44-3**, and **44-4** are coated on a common transparent structure **42**. This example is merely illustrative and the mirrors may have discrete respective transparent structures if desired. As yet another alternative, transparent structure **42** may be omitted (similar as discussed in connection with FIG. 3).

[0051] FIG. 6 is a side view of annular mirrors **44** and display module **14A** from FIG. 6. As shown in FIG. 6, the annular mirrors may be concentric, with each annular mirror having a central opening that overlaps display module **14A**.

[0052] For the annular mirrors described herein in connection with FIGS. 3 and 5, the curvature and/or angle of the mirror relative to the XZ-plane may remain constant around the circumference of the annular mirror. The example in FIG. 5 of including four discrete annular mirrors is merely illustrative. In general, any desired number of annular mirrors may be used (e.g., one, two, three, four, more than four, more than five, more than eight, etc.).

[0053] It is noted that, in FIG. 3, mirrors **44** and **48** as well as transparent structures **42** and **46** may collectively be referred to as collimating optics **34**. In FIG. 5, mirrors **44-1**, **44-2**, **44-3**, **44-4**, and **48** as well as transparent structure **42** may collectively be referred to as collimating optics **34**.

[0054] The mirrors described herein may have spherical curvature, aspherical curvature, or free form curvature.

[0055] The collimating optics of FIGS. 3 and 5 may advantageously be lightweight, compact, and have a low cost and complexity. The compact design may allow greater flexibility in the integration of the collimating optics into system **10**.

[0056] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A display system comprising: a waveguide having first and second opposing sides; a first mirror on the first side of the waveguide; a display module on the second side of the waveguide that produces image light; and a second mirror on the second side of the waveguide, wherein the second mirror has an annular footprint with a central opening and wherein the display module and the first

mirror overlap the central opening.

2. The display system defined in claim 1, wherein the image light produced by the display module passes through the waveguide from the second side towards the first side, then reflect off the first mirror, then passes through the waveguide from the first side towards the second side, and then reflects off the second mirror.

3. The display system defined in claim 1, further comprising: an annular input coupler on the first side of the waveguide, wherein the annular input coupler has an additional central opening and wherein the display module and the first mirror overlap the additional central opening.

4. The display system defined in claim 3, wherein the image light produced by the display module passes through the waveguide from the second side towards the first side, then reflects off the first mirror, then passes through the waveguide from the first side towards the second side, then reflects off the second mirror, then passes through the waveguide from the second side towards the first side, and then is redirected by the annular input coupler.

5. The display system defined in claim 3, wherein the annular input coupler comprises a surface relief grating.

6. The display system defined in claim 3, further comprising: an output coupler that redirects the image light towards an eye box on the second side of the waveguide.

7. The display system defined in claim 3, further comprising: an output coupler that redirects the image light towards an eye box on the first side of the waveguide.

8. The display system defined in claim 3, wherein the second mirror has a surface with concave curvature that faces the annular input coupler.

9. The display system defined in claim 1, wherein the waveguide has a temple region and a nasal region and wherein the display module and the first and second mirrors overlap the temple region.

10. The display system defined in claim 1, wherein the waveguide has a temple region and a nasal region and wherein the display module and the first and second mirrors overlap the nasal region.

11. The display system defined in claim 1, wherein the first mirror has a surface with convex curvature that faces the display module.

12. The display system defined in claim 1, further comprising: a third mirror on the second side of the waveguide, wherein the third mirror has an additional annular footprint with an additional central opening and wherein the third mirror is formed within the central opening of the second mirror.

13. The display system defined in claim 1, further comprising: an output coupler that redirects the image light towards an eye box on the first side of the waveguide.

14. The display system defined in claim 1, further comprising: an output coupler that redirects the image light towards an eye box on the second side of the waveguide.

15. A display system comprising: a waveguide having first and second opposing sides; a first mirror on the first side of the waveguide; a surface relief grating on the first side of the waveguide, wherein the surface relief grating has a first annular footprint with a first central opening and wherein the first mirror overlaps the first central opening; a display module on the second side of the waveguide that produces image light that passes through the first central opening to reach the first mirror; and a second mirror on the second side of the waveguide that receives the image light from the first mirror.

16. The display system defined in claim 15, wherein the second mirror has a second annular footprint with a second central opening and wherein the display module and the first mirror overlap the second central opening.

17. The display system defined in claim 15, wherein the surface relief grating redirects the image light from the second mirror to propagate by total internal reflection across the waveguide.

18. The display system defined in claim 15, wherein the first mirror has a convex surface that receives the image light from the display module.

19. The display system defined in claim 15, wherein the second mirror has a concave surface that

receives the image light from the first mirror.

20. A display system comprising: a waveguide having first and second opposing sides; a display module that produces image light; an input coupler on the waveguide that couples the image light into the waveguide; and collimating optics that collimate the image light from the display module for the input coupler, wherein the collimating optics comprise a first mirror on the first side of the waveguide and a second mirror on the second side of the waveguide.

21. The display system defined in claim 20, wherein the display module is positioned on the first side of the waveguide and wherein the input coupler is positioned on the second side of the waveguide.

22. The display system defined in claim 21, wherein the image light passes through a thickness of the waveguide three times between leaving the display module and reaching the input coupler.
